# Zerui Cheng

Princeton, NJ 08540, United States

zerui.cheng@princeton.edu • zerui-cheng.com • zerui-cheng

Marco-Cheng • ZeruiCheng • Google Scholar


## Education

**Princeton University, Dept of Electrical and Computer Engineering** — **Princeton, NJ, United States**
*Ph.D. Candidate, GPA: 4.00/4.00* — *Aug 2023 – Jul 2027 (expected)*

- Research focus: LLM evaluation, Synthetic data generation, Large language models, Mechanism design
- Ph.D. General exam passed and incidental M.A. degree received in May 2025. Advisor: Prof. Pramod Viswanath

**Tsinghua University, the Institute for Interdisciplinary Information Sciences** — **Beijing, China**
*B.Eng. in Computer Science (Honors), GPA: 3.91/4.00* — *Aug 2019 – Jun 2023*

- "Yao Class" talent program (founded by Turing Award winner Andrew C. Yao), graduated **summa cum laude**
- **"Yao Award"** recipient (most prestigious honor, top 10% of Yao Class)
- **Excellent diploma thesis award** winner (top 5% of Yao Class); Thesis Advisor: Prof. Zhixuan Fang
- Elected **Vice President** (2021-22) and **President** (2022-23) of **Yao Class Students' Congress**.
- Visiting student researcher at Duke University (2022), Advisor: Prof. Fan Zhang
- Visiting student researcher at the University of California, Berkeley (2022), Advisor: Prof. Dawn Song

## Professional Experiences

**Citadel Securities** — **Miami, FL**
*Quantitative Research Intern* — *Jun 2026 - Aug 2026 (expected)*

**Tencent Hy (HunYuan) Team** — **Shanghai, China**
*Research Intern (QingYun Program)* — *Mar 2026 - May 2026*
Building **long-horizon** benchmarks to measure **continual learning** abilities of AI agents.
Contributor to the evaluation of model `Hy3 Preview`.

**ByteDance Seed** — **Remote**
*Part-time Student Researcher* — *Sep 2025 – Jan 2026*
Research on **reliable and generalizable evaluation of large language models**.
First author of VeRA (synthetic reasoning data generation) and TabularMath (synthetic benchmark for tabular models).
Second author of the papaer on Generalization Spectrum (ICML 2026).
Contributor to the evaluation and coding data recipe of model `Seed 1.8`, `Seed 2.0 Pro`.
Mentor: Dr. Jiashuo Liu

**Surf AI, Cybertino Labs** — **Remote**
*Research Lead (Part-time)* — *Aug 2025 – Dec 2025*
Research on the intersection of crypto and AI, building benchmarks for real-life adversarial high-stakes scenarios.
Co-first author of CAIA (the first Crypto AI Agent benchmark). Surf AI raised $15 million in seed round.
Mentor: Ryan Li (CEO of Cybertino Labs)

**Kite AI** — **Remote**
*Research Collaborator (Part-time)* — *Feb 2025 – Oct 2025*
Research on Proof of Attributed Intelligence for AI-native blockchain operations.
Core author of Kite AI whitepaper: From Human-Centric to Agent-Native: Building Trustless Payment Infrastructure for Agentic AI. Kite AI raised $33 million in seed round and series A combined.
Mentor: Scott Shi (CTO of Kite AI)

**JQ Investments** — **Shanghai, China**
*Quantitative Research Intern* — *Aug 2024 – Sep 2024*
Developed alpha strategies through large-scale financial data analysis.
Mentor: Kevin Zhang

**Sentient Foundation** **Princeton, NJ**
*AI Fellow* *May 2024 – Aug 2024*
Led formulation and development of OML framework for AI intellectual property protection using cryptographic primitives.
First author of Sentient whitepaper "OML – Unleashing the Era of AI Entrepreneurship".
Sentient raised $85 million in seed round and secured over $1 billion in FDV after launch.
Mentor: Prof. Pramod Viswanath (co-founder of Sentient Foundation), Prof. Sewoong Oh

**Pond Crypto** **Remote**
*Research Engineering Lead (Non-executive)* *Apr 2023 – Dec 2024*
Poind raised $7.5 million in seed round. Mentor: Dylan Zhang

**PolyHedra Network** **Remote**
*Research Intern (Part-time)* *Aug 2022 – Dec 2022*
Designed blockchain cross-chain bridges using zero-knowledge proofs. Author of PolyHedra whitepaper (zkBridge).
Mentor: Dr. Tiancheng Xie (CTO of Polyhedra)

**Shanghai Qizhi Institute** **Shanghai, China**
*Research Intern* *Jul 2021 – Aug 2021*
Worked on modeling and queueing theory analysis of blockchain transaction waiting time.
Mentor: Prof. Zhixuan Fang

## Selected Publications

*For most recent updates, the full publication list is available at: Link to Google Scholar.*
∗ denotes equal contributions. † denotes equal advising.
**2026**:

- Jiayang Sun, Jiawei Xu, Maxm Pan, **Zerui Cheng**†, Pramod Viswanath†. "When Wrong Answers Look Right: Multi-Agent Debate for High-Precision Verification of Hard Competition Mathematics." (research preview, 2026).
- **Zerui Cheng**, Jiawei Xu, Maxm Pan. "Learning to Improve at Test Time: Long-Horizon Continual Learning Benchmarks for Language Agents" (research preview, 2026, with Tencent Hy team).
- Jinghan Zhang, **Zerui Cheng**, Shiqi Chen, Ge Zhang, Wenhao Huang, Jiashuo Liu, Junxian He, Tianle Cai. "The Generalization Spectrum: A Chromatographic Approach to Evaluating Learning Algorithms." (ICML 2026, with ByteDance Seed team)
- **FrontierCS** [Project Page]
  Qiuyang Mang, Wenhao Chai, Zhifei Li, Huanzhi Mao, Shang Zhou, Alexander Du, Hanchen Li, Shu Liu, Edwin Chen, Yichuan Wang, Xieting Chu, **Zerui Cheng**, Yuan Xu, Tian Xia, Zirui Wang, Tianneng Shi, Jianzhu Yao, Yilong Zhao, Qizheng Zhang, Charlie Ruan, Zeyu Shen, Kaiyuan Liu, Runyuan He, Dong Xing, Zerui Li, Zirong Zeng, Yige Jiang, Lufeng Cheng, Ziyi Zhao, Youran Sun, Wesley Zheng, Meiyuwang Zhang, Ruyi Ji, Xuechang Tu, Zihan Zheng, Zexing Chen, Kangyang Zhou, Zhaozi Wang, Jingbang Chen, Aleksandra Korolova, Peter Henderson, Pramod Viswanath, Vijay Ganesh, Saining Xie, Zhuang Liu, Dawn Song, Sewon Min, Ion Stoica, Joseph E. Gonzalez, Jingbo Shang, Alvin Cheung. "FrontierCS: Evolving Challenges for Evolving Intelligence." (ICML 2026)
- Canhui Chen∗, **Zerui Cheng**∗, Xinze Liu, Shutong Qu, Zhixuan Fang. "ValueMine: A Blockchain-based System for Permissionless and Trustless Computation." (IEEE Transactions on Networking, 2026)
- **VeRA** [Project Page]
  **Zerui Cheng**, Jiashuo Liu, Chunjie Wu, Jianzhu Yao, Pramod Viswanath, Ge Zhang, Wenhao Huang. "VeRA: Verified Reasoning Data Augmentation at Scale." (2026, with ByteDance Seed team).
- **TabularMath** [Project Page]
  **Zerui Cheng**, Jiashuo Liu, Jianzhu Yao, Pramod Viswanath, Ge Zhang, Wenhao Huang. "TabularMath: Evaluating Computational Extrapolation in Tabular Learning via Program-Verified Synthesis." (2026, with ByteDance Seed team).
- **FutureX Pro** [Project Page]
  Jiashuo Liu, Siyuan Chen, Zaiyuan Wang, Zhiyuan Zeng, Jiacheng Guo, Liang Hu, Lingyue Yin, Suozhi Huang, Wenxin Hao, Yang Yang, **Zerui Cheng**, Zixin Yao, Haoxin Liu, Jiayi Cheng, Yuzhen Li, Zezhong Ma, Bingjie Wang, Bingsen Qiu, Xiao Liu, Zeyang Zhang, Zijian Liu, Jinpeng Wang, Mingren Yin, Tianci He, Yali Liao, Yixiao Tian, Zhenwei Zhu, Anqi Dai, Ge Zhang, Jingkai Liu, Kaiyuan Zhang, Wenlong Wu, Xiang Gao, Xinjie Chen, Zhixin Yao, Zhoufutu Wen, B Aditya Prakash, Jose Blanchet, Mengdi Wang, Nian Si, Wenhao Huang. "FutureX-Pro: Extending Future Prediction to High-Value Vertical Domains." (2026, with ByteDance Seed team).
- Long Phan, ..., Lennart Finke, **Zerui Cheng**, Jennifer Zampese, ... et al. "A benchmark of expert-level academic questions to assess AI capabilities" [Nature Vol. 649, No. 8099 (2026): Page 1139-1146]

**2025**:

- **AutoCode** [Project Page]
  Shang Zhou*, Zihan Zheng*, Kaiyuan Liu*, Zeyu Shen*, **Zerui Cheng***, Zexing Chen, Hansen He et al. "AutoCode: LLMs as Problem Setters for Competitive Programming." [ICLR 2026].
- **CAIA (Crypto AI Agent Benchmark):** Zeshi Dai*, Zimo Peng*, **Zerui Cheng***, and Ryan Yihe Li. "When Hallucination Costs Millions: Benchmarking AI Agents in High-Stakes Adversarial Financial Markets." [AAAI 2026 AI-4-Finance, *Oral*]
- **TAO (Tolerance-Aware Optimistic Verification for Floating-Point Neural Networks):**
  Jianzhu Yao, Hongxu Su, Taobo Liao, **Zerui Cheng**, Huan Zhang, Xuechao Wang, and Pramod Viswanath. "Tolerance-Aware Optimistic Verification for Floating-Point Neural Networks." [EuroSys 2026].
- **PeerBench** [Project Page]
  **Zerui Cheng***, Stella Wohnig*, Ruchika Gupta*, Samiul Alam*, Tassallah Abdullahi, João Alves Ribeiro, Christian Nielsen-Garcia, Saif Mir, Siran Li, Jason Orender, Seyed Ali Bahrainian, Daniel Kirste, Aaron Gokaslan, Carsten Eickhoff, and Ruben Wolff. "Position: Benchmarking is Broken - Don't Let AI Be Its Own Judge" [NeurIPS 2025]
- **LiveCodeBench Pro** [Project Page]
  Zihan Zheng*, **Zerui Cheng***, Zeyu Shen*, Shang Zhou*, Kaiyuan Liu*, Hansen He*, Dongruixuan Li, Stanley Wei, Hangyi Hao, Jianzhu Yao, Peiyao Sheng, Zixuan Wang, Wenhao Chai, Aleksandra Korolova, Peter Henderson, Sanjeev Arora, Pramod Viswanath, Jingbo Shang, and Saining Xie. "LiveCodeBench Pro: How Do Olympiad Medalists Judge LLMs in Competitive Programming?" [NeurIPS 2025]
  - *Cited by* Google Gemini, Meta Muse Spark, ByteDance Seed*, etc. for performance reporting in their model cards.*
  - *Jun 24, 2025: Featured in MIT Technology Review.*
  - *Aug 4, 2025: Invited talk at OpenAGI Symposium at the University of California, Berkeley*
  - *Aug 15, 2025: Invited talk at "AI4Science x AI Security Community" hosted by alphaXiv*
  - *Dec 4, 2025: Invited talk at OpenAGI Symposium at NeurIPS 2025, San Diego*
- **OML Primitive: Zerui Cheng**, Edoardo Contente, Benjamin Finch, Oleg Aleksandrovich Golev, Jonathan Hayase, Andrew Miller et al. "OML: A Primitive for Reconciling Open Access with Owner Control in AI Model Distribution." In NeurIPS 2025 Lock-LLM Workshop: Prevent Unauthorized Knowledge Use from Large Language Models.
  - *Oct 15, 2025: Invited public talk hosted by Decentralized AI Institute and 5 other co-hosts.*
- **Kite AI Whitepaper** [Project Page]
  Scott Shi, **Zerui Cheng**, Chen Xi, Yi Huang, Lyon Li, Uddhav Marwaha, David Weber, and Chi Zhang. "From Human-Centric to Agent-Native: Building Trustless Payment Infrastructure for Agentic AI." (2025)
- **SPIN-Bench** [Project Page]
  Jianzhu Yao*, Kevin Wang*, Ryan Hsieh, Haisu Zhou, Tianqing Zou, **Zerui Cheng**, Zhangyang Wang, and Pramod Viswanath. "Spin-bench: How well do llms plan strategically and reason socially?" [COLM 2025]
- **HLE**: Long Phan, ..., Lennart Finke, **Zerui Cheng**, Jennifer Zampese, ... et al. "Humanity's Last Exam" (2025).

**2024**:

- **OML (Open, Monetizable, and Loyal AI): Zerui Cheng**, Edoardo Contente, Ben Finch, Oleg Golev, Jonathan Hayase, Andrew Miller, Niusha Moshrefi, Anshul Nasery, Sandeep Nailwal, Sewoong Oh, Himanshu Tyagi, and Pramod Viswanath. "Reclaiming "Open AI" – AI Model Serving Can Be Open Access, Yet Monetizable and Loyal" (2024).
  - *Sep 12, 2024: Initial release as the whitepaper of AI startup Sentient Foundation*
  - *Apr 11, 2025: Poster presentation at Citadel Securities PhD Summit 2025 in Miami*
  - *Jun 16, 2025: Invited talk at Prof. Matthias Bethge's research group at the University of Tübingen*

**2023**:

- **Sakshi:** Suma Bhat, Canhui Chen, **Zerui Cheng**, Zhixuan Fang, Ashwin Hebbar, Sreeram Kannan, Ranvir Rana, Peiyao Sheng, Himanshu Tyagi, Pramod Viswanath, and Xuechao Wang. "Sakshi: Decentralized AI Platforms" (2023).
- **PoCW (Proof of Crowdsourcing Work):** Canhui Chen*, **Zerui Cheng***, Shutong Qu, and Zhixuan Fang. "Crowdsourcing Work as Mining: A Decentralized Computation and Storage Paradigm". [APNET 2023]

**2022**:

- **zkBridge:** Tiancheng Xie, Jiaheng Zhang, **Zerui Cheng**, Fan Zhang, Yupeng Zhang, Yongzheng Jia, Dan Boneh, and Dawn Song. "zkBridge: Trustless Cross-chain Bridges Made Practical". [ACM CCS 2022]

## Talks and Presentations

**Invited talk at Huawei Noah's Ark Lab Seminar Series** — **Remote**
*1-hour talk on Verifiable Evaluation and Synthetic Data in Code and Math for Reasoning AI Agents.* — *Jun 2026*

**Invited talk at the 3rd Universal Cup Conference, Huawei Lianqiu Lake R&D Center** — **Shanghai, China**
*30-min talk on AI for Competitive Programming (delivered by Stanley Wei).* — *May 2026*

**Invited talk at OpenAGI Symposium at NeurIPS** — **San Diego, CA**
*20-min talk on Open-Source AI for Competitive Programming.* — *Dec 2025*

**Invited public talk hosted by Decentralized AI Institute and 5 other co-hosts.** — **Online**
*1-hour talk on OML Primitive. [Link to video recording]* — *Oct 2025*

**Invited talk at "AI4Science x AI Security Community" hosted by alphaXiv** — **Online**
*1-hour talk on LiveCodebench Pro (w/ Peiyao Sheng).* — *Aug 2025*

**Invited talk at OpenAGI Symposium hosted by Sentient** — **Berkeley, CA**
*20-min talk on LiveCodebench Pro at UC Berkeley.* — *Aug 2025*

**Invited talk at the University of Tübingen hosted by Prof. Matthias Bethge** — **Online**
*1-hr talk on OML.* — *Jun 2025*

**Ph.D. General Exam Research Seminar** — **Princeton, NJ**
*1-hr talk on OML. Committee members: Prof. Pramod Viswanath, Prof. Sanjeev Kulkarni, Prof. Chi Jin* — *Apr 2025*

**Poster presentation at Citadel Securities PhD Summit** — **Miami, FL**
*2-hr poster session with top PhD students and quantitative researchers at CitSec on OML.* — *Apr 2025*

**Poster presentation at Princeton DeCenter Spring Conference** — **Princeton, NJ**
*1-hr poster session on SCTest: Automated smart contract generation with AI (w/ group members).* — *Apr 2024*

**Invited talk at Yao Seminar #48** — **Beijing, China**
*1-hr talk on zkBridge: trustless cross-chain bridges by zero-knowledge proofs.* — *Jun 2023*

**Presentation at IC3 Blockchain Summer Camp 2022** — **Ithaca, NY**
*10-min lightning session on multi-agent selfish mining (w/ Shutong Qu) at Cornell University.* — *Nov 2022*

## Scholarships and Academic Awards

**NeurIPS 2025 Scholar Award** — **NeurIPS 2025 Next Generation & Accessibility Chairs**
*Dec 2025*

**Travel Award: Toby and Jack Wolf Fund for Excellence** — **Princeton University**
*Oct 2025*

**First-year Ph.D. Fellowship** — **Princeton University**
*Aug 2023*

**Departmental summa cum laude** — **Tsinghua University**
*Top 20% in Yao Class* — *Jun 2023*

**Excellent Diploma Thesis Award** — **Tsinghua University**
*Top 5% in Yao Class* — *Jun 2023*

**Yao Award (Bronze Medal)** — **Tsinghua University**
*Top 10% in Yao Class* — *Sep 2022*
Most prestigious honor in Yao Class, awarded by Dean Andrew Chi-Chih Yao

**Scholarship for Comprehensive Excellence** — **Tsinghua University**
*2022*

**Scholarship for Excellence in Academic Achievements** — **Tsinghua University**
*2021*

**Scholarship for Excellence in Scientific and Technological Innovation** — **Tsinghua University**
*2021*

**Scholarship for Excellence in Social Activities and Voluntary Work** — **Tsinghua University**
*2020*

**Xuetang Scholarship** — **Tsinghua University**
*Full tuition exemption* — *2019-2023*

## Competitive Programming Achievements

### Career Achievement

Competitive Programming **Hall of Fame Member** [Hall of Fame Personal Page]

### Recent Highlights in World-class Competitions

**May 2026**: **Outstanding Award, 9th worldwide**, Huawei Challenge at the 3rd Universal Cup Finals

**May 2026**: **Finalist, 22nd worldwide**, The 3rd Universal Cup Finals

### Recent Highlights in North America

**May 2025**: **26th Place**, ICPC North America Championship (NAC 2025)

**Nov 2024**: **4th Place**, ICPC Greater New York Regional (qualified for NAC 2025 representing Princeton)

**Nov 2024**: **93rd worldwide** (top 0.4%), Meta HackerCup

**May 2024**: **22nd Place**, ICPC North America Championship (first solution award)

**Nov 2023**: **3rd Place**, ICPC Greater New York Regional (qualified for NAC 2024 representing Princeton)

### University Competitions

**Dec 2024**: **2nd Place**, Princeton Computer Science Contest (Princeton COSCON 2024)

**May 2024**: **Finalist** (Top 16 in USA), MIT Informatics Tournament Spring Invitational (MITIT 2024)

**Dec 2023**: **1st Place**, Princeton Computer Science Contest (Princeton COSCON 2023)

### Earlier Achievements in East Asia

**2020-2022**: **Multiple Gold Medals** in ICPC/CCPC regional contests and continental finals.
ICPC: Gold (Yinchuan Regional 2021), Silver (East-Asia Continental Final 2021);
CCPC: Gold (Changchun 2020, Harbin 2021), Silver (Mianyang 2022);

**2019**: **2nd Place**, LeetCode Cup National Programming Contest

**2017-2018**: **Silver Medalist** (2x), Chinese National Olympiad in Informatics (NOI).

### Problem Setting

**Jul 2021**: Problem setter of Chinese National Olympiad in Informatics (NOI 2021 Day 2 Problem 1).

**2021-2022**: Problem setter and validator of Tsinghua University Programming Contest (THUPC 2021,2022).

**Nov 2017**: Lead author of Codeforces Round 447.

## Other Honors

**Aug 2022**: **1st Place**, IC3 Blockchain Summer Camp Hackathon 2022

**Dec 2020**: **First Prize**, China Collegiate Mathematics Contest

## Teaching and Service

**Princeton ECE 399: Junior Independent Work**
*Graduate Research Mentor* — *Feb 2026 – now*
Mentee: Jiayang Sun (exchange student at Princeton, top 1% at HKU)

- Mentor and advisor of the research paper "When Wrong Answers Look Right: Multi-Agent Debate for High-Precision Verification of Hard Competition Mathematics".

**Princeton COS/ECE 470: Principles of Blockchains**
*Head Teaching Assistant* — *Sep 2024 – Jan 2025*

- Redesigned final project combining blockchain and AI, receiving exceptional student feedback
- Led 3 hands-on in-class labs and coordinated grading for 80+ students
- Nominated for Princeton E-Council Teaching Awards (Fall 2024)

**The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)**
*Invited Reviewer (Position Track)* — *2026*

**Forty-Third International Conference on Machine Learning (ICML 2026)**
*Invited Reviewer (Main Track)* — *2026*

**ACM Transactions on Multimedia Computing, Communications and Applications (ACM TOMM)**
*Invited Reviewer* — *2026*

**The Fourteenth International Conference on Learning Representations (ICLR 2026)**
*Invited Reviewer* — *2025*

**NeurIPS 2025 Reliable ML Workshop**
*Invited Reviewer* — *2025*

**IEEE/ACM Transactions on Networking (IEEE ToN)**
*Invited Reviewer* — *2025*

**IEEE Journal on Selected Areas in Communications (IEEE JSAC)**
*Invited Reviewer* — *2022*

**China Computer Federation (CCF)**
*Student Expert and Problem Setter of National Olympiad in Informatics of China (NOI 2021)* — *2021*
Authored problem ("Quantum Communication", Day 2 Problem 1) voted "best in contest"

**Tsinghua University Algorithm Association**
*Core contributor.* — *2020-22*

**Yao Class Youth League**
*Deputy Secretary.* — *2021-22*

**Yao Class Students' Congress**
*President (2022-23); Vice president (2021-22); Member (2020-21)* — *2020-23*

## Media Coverage

**Article "Que valent les comparateurs d'IA ?"** — **Sciences et Avenir**
*Interviewed by journalist Théo Brajard on PeerBench paper and the philosophy of AI Evaluation.* — *Jun 2026*

**Article "Plongée dans la fabrique chinoise des futurs génies de l'IA"** — **Le Figaro**
*Interviewed by journalist Suzanne Duroy on personal experience and research.* — *May 2026*

**Article "Can we fix AI's evaluation crisis?"** — **MIT Technology Review**
*Covered our research paper LiveCodeBench Pro.* — *Jun 2025*

**TV Show "Super Brain Season 10"** — **JSBC (Jiangsu Satellite TV)**
*One of the top 50 finalists invited for onsite filming.* — *Jan 2023*

## Languages

**Chinese (Mandarin)**: Native
**English**: Fluent (TOEFL iBT: 111/120)